

WHARTEXA™ MATERIAL SAFETY DATA SHEET (MSDS)

SECTION 1: Identification

1.1 Product Identifier

Product Name: WHARTEXA™

1.2 Product Description

Human Umbilical Cord MSC-Derived Extracellular Vesicles

Frozen sterile suspension containing purified extracellular vesicles intended for professional cosmetic application.

1.3 Other Means of Identification

Strength: 20 & 60 Billion Bioactive Particles

Fill Volume: 5 mL

1.4 Recommended Use

Professional cosmetic use for:

scalp biologic support

vital skin areas

sensitive external cosmetic regions

1.5 Supplier Details

Manufacturer / Supplier: Nanoventra

Address:

1441 West Innovation Way, Suite 400

Lehi, Utah 84043

United States of America

Email:

nanoventra@gmail.com

1.6 Emergency Contact

Available through supplier documentation.

SECTION 2: Hazard Identification

2.1 Classification

Physical hazard: Not classified

Health hazard: Not classified under intended cosmetic use

Environmental hazard: Not classified

2.2 Label Elements

Hazard pictogram: Not applicable

Signal word: Not applicable

Hazard statement: Not applicable

2.3 Other Hazards

No known significant hazard under intended professional cosmetic handling.

SECTION 3: Composition / Information on Ingredients

3.1 Main Active Substance

Human Umbilical Cord MSC-Derived Extracellular Vesicles

3.2 Biological Source

Derived from Wharton's Jelly human umbilical cord mesenchymal stem cells

3.3 Naturally Present Biological Fractions

- extracellular vesicles
- microRNAs
- cytokine-associated signaling molecules
- membrane proteins
- growth-related signaling factors

3.4 Additional Components

Preservative system only

3.5 Excluded Additives

- no peptides
- no hyaluronic acid
- no synthetic growth factors
- no cosmetic fillers

SECTION 4: First Aid Measures

4.1 Eye Contact

Rinse with clean water for several minutes if accidental exposure occurs.

4.2 Skin Contact

Safe under intended external cosmetic use.

4.3 Inhalation

Move to fresh air if aerosol exposure occurs.

4.4 Ingestion

Seek medical advice.

SECTION 5: Fire Fighting Measures

5.1 Suitable Extinguishing Media

Water spray, dry chemical powder, CO₂

5.2 Specific Hazards

Biological material may degrade under excessive heat.

5.3 Protective Measures

Standard fire protection equipment.

SECTION 6: Accidental Release Measures

6.1 Personal Precautions

Wear gloves and protective clothing.

6.2 Environmental Precautions

Prevent discharge into drainage systems.

6.3 Cleaning Method

Absorb using sterile absorbent material.

SECTION 7: Handling and Storage

7.1 Safe Handling

Professional sterile handling only.

7.2 Storage Conditions

Store at **-20°C**

7.3 Storage Precautions

Avoid repeated freeze-thaw cycles.

SECTION 8: Exposure Controls / Personal Protection

8.1 Respiratory Protection

Not normally required under intended use.

8.2 Eye Protection

Protective eyewear during handling.

8.3 Hand Protection

Medical gloves recommended.

SECTION 9: Physical and Chemical Properties

9.1 Appearance

Clear to slightly opalescent suspension

9.2 Color

Light off-white to pale yellowish

9.3 Odor

Mild biologically neutral odor

9.4 pH

6.8 – 7.4

SECTION 10: Stability and Reactivity

10.1 Stability

Stable under frozen storage.

10.2 Conditions to Avoid

Heat, direct sunlight, repeated thawing.

10.3 Incompatible Materials

Strong oxidizing agents.

SECTION 11: Toxicological Information

11.1 Acute Toxicity

No known acute toxicity under intended professional cosmetic use.

11.2 Skin Irritation / Corrosion

Not expected to cause irritation under normal external cosmetic application.

11.3 Eye Irritation / Damage

No expected significant irritation under accidental external exposure. Avoid direct ocular contact.

11.4 Respiratory Sensitization

No known respiratory sensitization under normal handling conditions.

11.5 Skin Sensitization

No known sensitization effects reported under intended use.

11.6 Mutagenicity / Carcinogenicity

No known mutagenic or carcinogenic effects based on current biological composition.

11.7 Reproductive Toxicity / STOT (Single & Repeated Exposure)

No known reproductive toxicity or specific target organ toxicity under intended cosmetic use.

SECTION 12: Ecological Information

12.1 Environmental Impact

No specific ecological hazard identified.

12.2 Persistence

Biologic material expected to degrade naturally.

SECTION 13: Disposal Considerations

13.1 Disposal Method

Dispose of contents in accordance with applicable local, regional, and national regulations governing biological and clinical waste.

- Treat product as biologically derived material.
- Do not discharge into drains, surface water, or soil.
- **For small quantities:**
 - Absorb with suitable inert material (e.g., medical absorbent pads)
 - Place in designated biohazard waste container
- **For larger quantities:**
 - Collect using appropriate containment methods
 - Dispose through licensed medical or biological waste disposal services

SECTION 13: Disposal Considerations

Handling personnel should follow standard biosafety procedures and use appropriate PPE during disposal.

13.2 Packaging Disposal

Dispose of packaging materials (vials, caps, containers) in accordance with applicable waste management regulations.

- Empty containers may retain biological residues; treat as biohazard waste
 - Do not reuse packaging materials
 - **Dispose via:**
 - approved clinical waste streams, or
 - regulated sharps / sterile container disposal systems, where applicable
- Ensure that disposal is carried out in compliance with institutional, environmental, and regulatory guidelines.

SECTION 14: Transport Information

14.1 Shipping Method

Product must be transported under dry ice conditions to maintain its frozen biological stability. Dry ice ensures preservation of extracellular vesicle integrity during transit.

14.2 Transport Condition

Maintain a continuous frozen cold chain (-20°C or below) throughout transportation. Avoid any thawing events, as temperature fluctuation may compromise product quality.

14.3 Packaging

Product is supplied in a sealed sterile vial designed to maintain integrity and prevent contamination. Secondary insulated packaging with dry ice is required for safe shipment.

14.4 Handling During Transport

Handle with care to avoid physical shock, leakage, or vial damage. Ensure proper labeling as temperature-sensitive biological material and follow standard transport safety guidelines.

SECTION 15: Regulatory Information

15.1 Product Classification

Professional cosmetic use only

15.2 Medical Statement

Not intended to diagnose, treat, cure, or prevent disease

SECTION 16: Other Information

16.1 Batch Number

WHX-0426

16.2 Revision

Version 3.2

16.3 Prepared By

Nanoventra Technical Documentation Unit

16.4 Approval

Authorized internal documentation draft